

EduGenie – Backend Development

Project: EduGenie – Google Gemini Powered Learning Assistant

1. Introduction

The EduGenie backend is responsible for receiving requests from the frontend, processing them, communicating with the required learning modules, and returning results to the user.

2. Backend Technology

The backend is developed using:

- Python
- FastAPI
- Uvicorn
- Jinja2

3. Main Backend File

main.py

The `main.py` file acts as the main entry point of the FastAPI application. It initializes the application and manages the required routes and web requests.

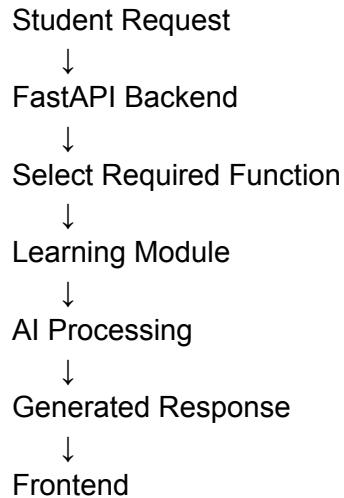
4. Learning Modules

The backend uses separate Python modules for different learning functions:

- `qna.py` – Question answering
- `explanation_module.py` – Simplified explanations
- `quiz_module.py` – Quiz generation
- `summary_module.py` – Text summarization
- `learning_path.py` – Learning recommendations

5. Request Processing

The basic backend process is:



6. API Endpoints

The application provides endpoints for the major learning functions, such as:

- `/qa`
- `/explain`
- `/quiz`
- `/summarize`
- `/learn/recommendations`

7. Error Handling

The backend handles common errors such as invalid requests, missing configuration, and AI service failures, and provides an appropriate response to the user.

8. Conclusion

The FastAPI backend provides the main processing layer of EduGenie and connects the web interface with the learning modules and AI service.